

SHANIYA JARRETT

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Objective: Astrophysics PhD student with aerospace industry experience, now focused on improving life here on Earth. Specializing in data science and machine learning to enhance how humans live, learn, and create. Skilled in modeling, analysis, and working with large-scale, multimodal datasets. **Seeking a part-time or summer internship** in related fields.

Education

University of Maryland 2024 - Present
Ph.D. Astrophysics & Astronomy
Fisk University 2022 - 2024
M.S. Physics
University of Colorado, Boulder 2014 - 2020
B.A. Astrophysics

Select Awards & Fellowships

• Outstanding Teaching Assistant Award 2025
• Margaret Cuningham Women's Center Leadership Award 2024
• Computer, Mathematical, and Natural Sciences
Exceptional Recruitment Award 2024
• National Science Foundation Graduate Research Fellow 2024
• Heising-Simons Foundation Mini-Grant Winner 2023

Skills

- **Hard Skills:** Proficient - Python, SQL, SciPy, Scikit-learn, Pandas, TensorFlow, Huggingface, Transformers, CNNs, Git/GitHub, Atlassian Tools, SolidWorks, Zemax, LaTeX; Familiar - AWS, APIs, Tableau
- **Soft Skills:** Cross-team Communication, Collaboration, Problem Solving, Adaptability, Initiative, Creativity, Research Mindset, Strategic Thinking, Project Management, Receptive to Feedback

Relevant Experience

Scientific Researcher

NASA Goddard Space Flight Center, University of Maryland, Vanderbilt University 2022 - Present

- Built and fine-tuned transformer-based neural networks on 40,000+ simulated time-series datasets using GPU acceleration, improving signal-classification performance on noisy astrophysical data.
- Leveraged pre-trained audio transformers (Hugging Face) to develop an interpretable ML pipeline, enabling transparent visualization and insight into model decisions.
- Explored multimodal fusion techniques combining time-series and image representations for improving classification performance.
- Applied spatial correlation and clustering analysis to 1 TB of simulated data to uncover feature relationships that enhanced predictive modeling of astrophysical event rates.

Laboratory of Atmospheric and Space Physics (LASP) 2017 - 2020

- Modeled digital micromirror devices using ray-tracing simulations to evaluate and optimize satellite imaging performance, reducing projected distortion by 15%
- Applied statistical and signal-processing techniques to multiple spacecraft datasets for anomaly detection and trend analysis, improving data reliability across missions.

Satellite Test Engineer, General Atomics, Centennial, CO 2020 - 2022

- Led vibration, mass-properties, and structural testing for the \$65 million *Gazelle* spacecraft
- Tested 5+ major subsystems across hardware and software, and co-pioneered the first thermal-vacuum campaign, establishing spacecraft-qualification standards.
- Co-authored multiple test procedures that identified recurring anomalies, improving data reliability and reducing mission risk while collaborating with cross-functional teams and presenting results to stakeholders.

Outreach Projects

- *Program Manager/Creator*, AstroBeats Project (see astrobeats.us for music) 2023 - 2024
 - Designed a workshop where middle school girls turned astrophysical data into music; awarded a mini-grant and leadership honor; recognized by Vanderbilt University and multiple podcasts for STEM innovation
- *Instructor*, Black Girls Becoming (BGB) Summer Institute 2023
 - Developed and taught an astrophysics curriculum as the sole student instructor among Vanderbilt faculty for 20+ middle school girls, providing their first hands-on exposure to STEM.
- *Workshop Guide*, Foundations of Astronomical Data Science 2025
 - Helped guide participants to learn Python, ADQL, and SQL to build skills for research

3 Publications | 6+ Conference Presentations | 3 Invited Talks | 10+ Volunteering Experiences | 11+ Awards